

The Econometrics of Kinked Budget Constraints

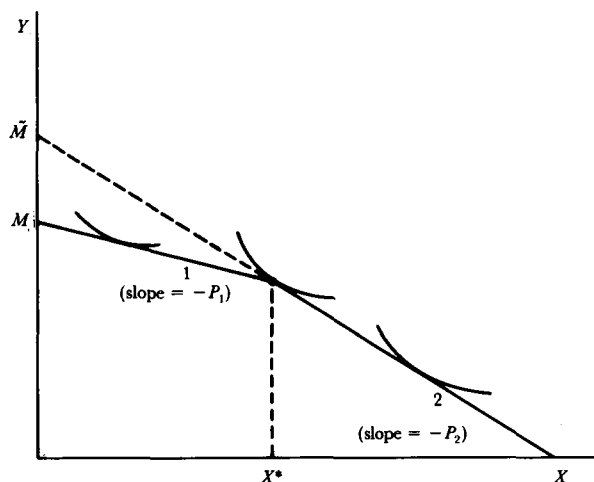
Robert Moffitt

In the last several years a branch of applied econometrics has developed which is devoted to the development of techniques for the estimation of demand and other functions when the budget constraint is “piecewise-linear,” or “kinky”—that is, when the constraint consists of a number of segments joined together at kink points. Such constraints most frequently arise from government tax and transfer programs. For example, the federal income tax imposes a piecewise-linear schedule by the use of constant marginal tax rates within brackets, a problem analyzed by Hausman (1981). The payroll tax in the Social Security system likewise generates a kinked schedule inasmuch as the marginal tax rate on earnings is zero above a certain earnings maximum. Federal and state transfer programs for the poor result in piecewise-linear constraints because transfer benefits are “taxed” (in the sense of a negative income tax) as income rises and because that “tax rate” falls to zero at the kink point where an individual or family loses eligibility for the benefit altogether. Many government transfer programs impose additional nonlinearities by altering this tax rate by discrete amounts over the range of positive benefits as well. But kinked constraints sometimes arise in nongovernment contexts, a well-known example being the block pricing schedule commonly set by utilities and volume discounting in general.

Kinked budget constraints create two difficulties that have several interesting facets. One difficulty is that changes in tax and transfer schedules can have unexpected effects that can be exactly the opposite in sign to those expected from economic theory. Examples of this phenomenon are given below. The second difficulty, that with which this paper is mostly concerned, relates to the implications of kinked budget

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Figure 1

Convex budget set

constraints for the estimation of demand functions. Such constraints make estimation quite difficult for reasons that are often closely related to economic theory and to the nature of economic behavior of individuals, as noted in technical surveys by Hausman (1985) and Moffitt (1986a).

Demand Functions in the Presence of Kinked Constraints

Figure 1 shows a two-segment convex budget set for a two-good consumer choice problem. The price of Y is normalized to 1 and the price of X is P_1 up to X^* but increases to P_2 thereafter. Consumer income is M and the budget constraint is

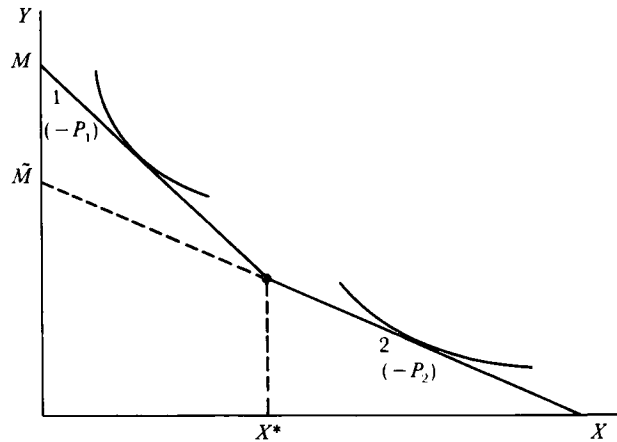
$$\begin{aligned}
 M &= P_1 X + Y \quad \text{if } X \leq X^* \\
 \tilde{M} &= P_2 X + Y \quad \text{if } X > X^*
 \end{aligned}
 \tag{1}$$

where $\tilde{M} = M + (P_2 - P_1)X^*$. $\tilde{M}$ is usually termed “virtual” income (Burtless and Hausman, 1978), and can be seen from Figure 1 to be the intercept of segment 2 if it is extended to the vertical axis. The consumer picks segment 1, segment 2, or the kink, depending upon which yields the highest utility.

An example of such a constraint arises in the federal estate tax, where bequests are taxable only if in excess of a certain amount (\$600,000 in 1986). If X is the amount of the bequest and if Y is amount of consumption of other goods, then the “price” of leaving a bequest is equal to 1 up to a certain amount—a dollar more of a

Figure 2

Non-convex budget set



bequest lowers consumption of other goods by the donor by one dollar. But every additional dollar of bequest above the exempt amount is taxed, and hence the “price” of leaving that dollar has increased.

Figure 2 shows a nonconvex budget set. The price declines from P_1 on segment 1 to P_2 on segment 2. The budget constraint can still be written as in equation (1). However, the kink will never be chosen as a utility maximum as long as the utility function is differentiable. Consequently, there are effectively only two choices in this case, segment 1 or segment 2. As one might expect, the fact that local maximums are not necessarily global maximums will result in special problems for the nonconvex case.

Nonconvex constraints are created, for example, by volume or quantity discounts. Retailers who offer low prices for buyers who purchase more than some minimum quantity of merchandise create for the buyer a lower price of the good for purchases above the minimum.

The comparative statics of consumer demand as enshrined in textbooks—most notably, the negative effect of price on quantity demanded under well-known conditions and the presumed positive effect of income on the quantities demanded of most goods—holds up in the presence of kinked budget sets only “within” segments. If the price or income change in question causes the consumer to change segments or kinks, there are, in general, no unambiguous directions of predicted effect.

Figures 3a–3d illustrate this point by showing several examples of the effects of changes in existing U.S. tax, transfer, and subsidy programs, each change inducing unexpected effects arising from the piecewise-linearity of the schedule in question. Figures 3a and 3b illustrate the choice of labor supply in the presence of a progressive income tax. Let the hourly wage rate be W and the marginal tax rates in the first and second brackets be t_1 and t_2 , ($t_1 < t_2$). Let the vertical axis represent take-home

Figure 3(a)
Decrease in t_1

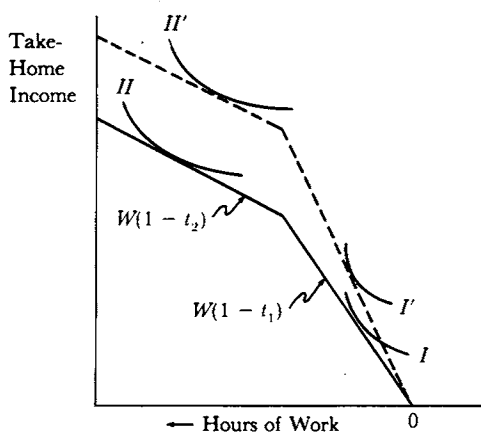


Figure 3(b)
Increase in t_1

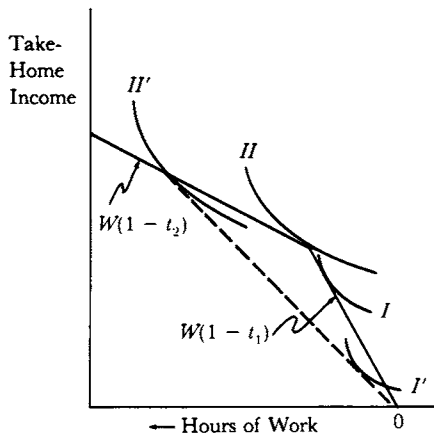


Figure 3(c)
Increase in s to 1

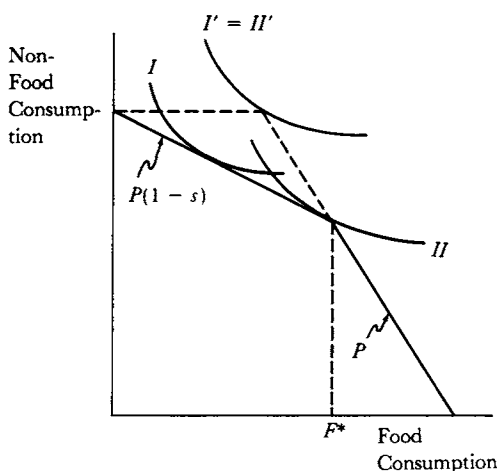
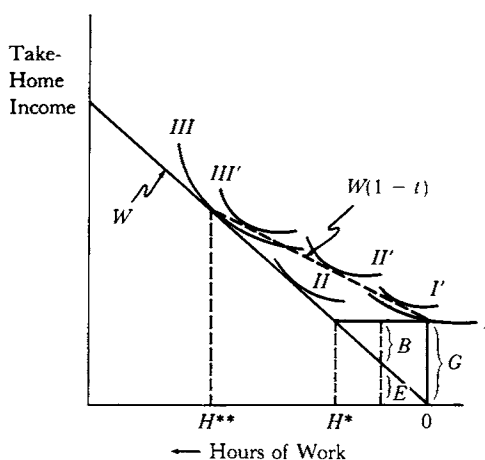


Figure 3(d)
Negative income tax



income and let the horizontal axis measure hours of leisure, so that hours of work start at zero at the right-most point of the constraint and then increase as one moves to the left. The budget constraint that results is of the shape shown by the solid lines in the two figures. The net wage rate (i.e., the increase in take-home income per hour of work) is $W(1 - t_1)$ for the first few hours of work beyond zero—moving up the constraint from the right-most point to the left—and then decreases to $W(1 - t_2)$ when the worker gets bumped into a higher tax bracket.

Figure 3a shows the effect of a decrease in t_1 , the marginal tax rate in the first bracket. As illustrated by the indifference curves shown, the individual in the first segment increases hours of work (I to I') as a result of the increase in the net wage rate. But the individual in the second bracket has only a change in income and no change in the net wage rate, so this individual reduces hours of work assuming leisure

is a normal good (II to II'). Thus the net effect of the tax-rate reduction is ambiguous in sign. Figure 3b shows an increase in t_1 , but in this case allowing the location of the kink point (i.e., the level of earnings at which the brackets switch) to change. In this case, the individual on the first segment reduces hours of work because of the tax-rate increase (I to I') but a different individual, one who was initially located at the kink point, increases hours of work because maximum utility "sticks" at the kink (II to II'). Once again, the effects of the tax change are ambiguous in sign. Clearly, changing tax rates in a progressive income tax system does not have simple effects.

Figure 3c shows the case of a food subsidy program. The horizontal axis measures food consumption and the vertical axis measures non-food consumption. In the figure, a food subsidy program is assumed to create the two-segment constraint shown by the solid lines initially. As food consumption rises from zero and the consumer moves down the first segment from left to right, starting at the vertical axis, the price of food is $P(1 - s)$, where P is the market price of food and s is the government subsidy rate. For example, if s is 0.50, the price of food is reduced by half. In practice, this can be implemented by permitting families to buy \$20 of food coupons from the government for \$10, or the same fraction of other amounts. But if the government only permits this subsidy up to a maximum amount of food consumption—shown by F^* in the figure—the price of food increases to the market price, P , for any consumption above that amount.

The dotted line in the figure shows the effect of converting this program to what is called a "take-it-or-leave-it" program—families are simply offered a certain amount of food coupons for free. This alters the constraint to that shown by the dotted line in the figure. For some individuals, this change will encourage additional food consumption (I to I'), but for those initially at the kink, it will reduce it (II to II'). Thus the net effect is ambiguous in sign, and the move to a seemingly more generous food subsidy program could actually reduce food expenditures among the poor. A change in the U.S. Food Stamp program of almost exactly this type occurred in 1977.

Figure 3d shows the case of a negative income tax and its effect on work incentives. The axes are the same as in Figures 3a and 3b. In the absence of any welfare program at all, the budget constraint is shown as a solid line running upward to the left from zero hours of work at the rate W , the hourly wage rate. The solid line running vertically to G and then horizontally leftward would be created by a welfare program that provides \$ G of benefits for an individual who does not work but which then reduces the welfare benefit by one dollar for every dollar that the individual earns. As the figure illustrates, if hours of work rise from zero, earnings (E) rise but the benefit (B) falls by the same amount, leaving total take-home income unchanged.

This type of welfare program has long been criticized by economists for its seeming work disincentives, for the welfare recipient initially at zero hours of work has no financial incentive to work more hours. Instead, if the benefit were reduced by some amount less than a dollar for every dollar of earnings—say by t —the dotted line shown in the figure would result. Along the dotted line, the net wage rate is $W(1 - t)$ instead of zero, as it was along the horizontal solid line. However, while this change will induce additional work effort on the part of those who initially were recipients (I

to I'), individuals who were initially non-recipients may reduce their hours of work to join the program (II to II' and III to III'). The problem arises because the kink point rises from H^* to H^{**} , and hence more individuals are eligible for benefits. The net effect of attempting to provide work incentives is, therefore, ambiguous in sign. Indeed, the existing empirical evidence suggests that a negative income tax in the U.S. would have essentially no effect on labor supply, if not a deleterious effect (Moffitt, 1986b).¹

These examples should suffice to illustrate not only the policy importance of the problem but also the type of econometric analysis necessary to properly forecast the effects of policy changes. For example, clearly it is necessary to estimate statistically not only "marginal" price and income elasticities—i.e., those that apply if individuals remain on the same segment—but also to determine when and how individuals shift segments and kinks. Also, the distribution of individuals over the constraint needs to be known because different persons in different initial locations react to changes in parts of the constraint in very different ways, often opposite in sign to one another. Indeed, the net effect of a policy change may very well depend critically upon the relative numbers of individuals located at different points.

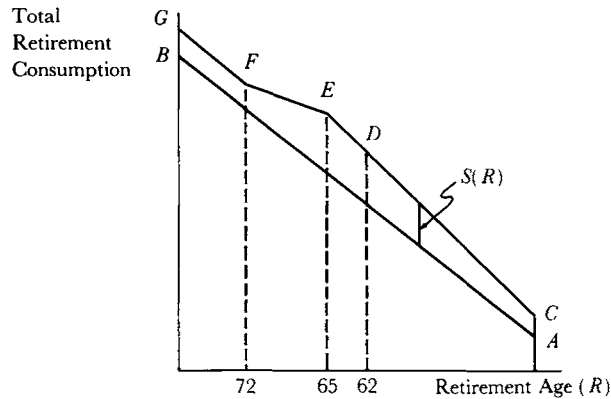
An Example from the Literature

As noted in the introduction, there have been many studies of the response to kinked budget constraints and applications to many different policy issues (see Hausman, 1985, and Moffitt, 1986a, for reviews). Most of the work on the subject has been in the area of labor supply and has been concerned with the effect of transfer programs such as an experimental negative income tax, the Aid to Families with Dependent Children (AFDC) program, Food Stamps, and social insurance programs such as Social Security, unemployment insurance, and disability insurance. Considerable work has also been addressed to the tax side, most of which has concerned the effects of income taxes on labor supply (Hausman, 1981). The effects of the income tax on charitable deductions and on housing expenditures have been studied as well. Finally, another area where the techniques have found application is in the study of federal and state grants-in-aid, for grant formulas are frequently nonlinear and induce behavior on the part of the recipient governmental unit that is correspondingly nonlinear.

The studies of Burtless and Moffitt (1984, 1985) of the effect of Social Security on retirement illustrate the types of data analysis issues that can arise in studies of kinked budget constraints. Their study was an econometric examination of the effect of the U.S. Social Security system on two variables—the age of retirement, and the hours of work of individuals after they had retired. The Social Security system creates a kinked budget constraint for both choices, as shown in Figures 4 and 5. In Figure 4, the lifetime budget constraint is AB in the absence of Social Security—retirement at an

¹This issue is discussed in the articles by Burtless and Gueron in the Winter 1990 issue of this journal.

Figure 4
Lifetime budget constraint

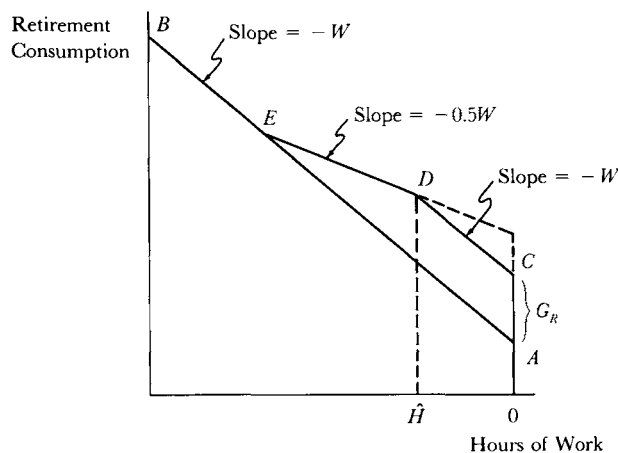


Source: Burtless and Moffitt (1985)

early age would leave total assets of $\$A$ for all consumption over the remaining lifetime, but the amount increases with retirement age assuming saving takes place. In the presence of Social Security the constraint is $CDEFG$, and the vertical difference between the two represents the lifetime value of Social Security benefits, $S(R)$. As retirement age, R , increases initially, $S(R)$ goes up because retirement benefits are increased at a disproportionate rate as lifetime earnings accumulate. But there is a sharp convex kink at age 65 because the Social Security benefit formula does not reward work beyond that age at an “actuarially fair” rate—although the retirement benefit is increased for retirement after age 65, it is not increased enough to compensate for the shorter length of life.

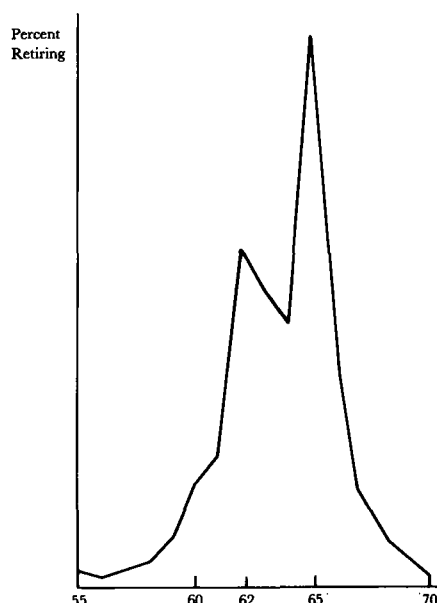
Figure 5 shows the budget constraint facing retirees who must choose their hours of work. The budget constraint in the absence of Social Security is line AB , with slope equal to the negative of the hourly wage rate (W). A Social Security benefit of G_R (it

Figure 5
Retirement hours of work constraint



Source: Burtless and Moffitt (1985)

Figure 6
Distribution of retirement ages



Source: Burtless and Moffitt (1984)

depends upon retirement age, R) is paid to those who do not work at all or who earn below a level of earnings termed the “exempt amount,” which is shown in the figure to be reached if the individual works $\hat{H}$ hours. For each dollar of earnings in excess of the exempt amount, the Social Security system reduces the benefit by \$0.50, resulting in a net wage of $0.5W$ over that range. Eventually benefits are exhausted entirely, at point E .

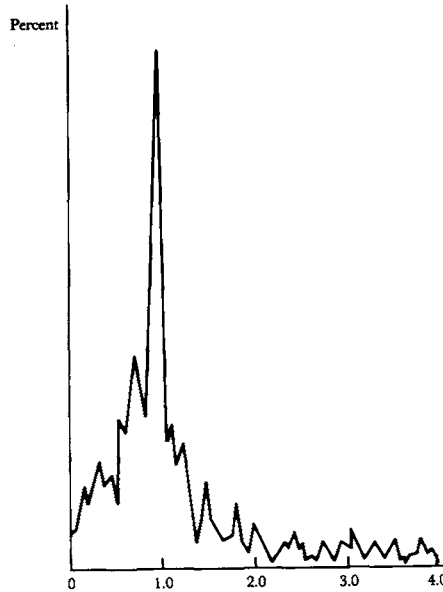
Burtless and Moffitt used a data set on older workers in the U.S. over the 1960s and 1970s called the Retirement History Survey, which allowed them to measure retirement ages and retirement hours of work for a sample of men. Figure 6 shows the distribution of retirement ages in their sample. The figure shows a marked peak of the distribution at age 65, the convex kink point of the lifetime budget constraint. Thus strong clustering around the kink was found.² Figure 7 shows the distribution of retirement earnings in the data relative to the Social Security exempt amount. Once again, there is strong visual evidence of clustering around the convex kink point of the constraint, indicating that individuals are quite aware of the point at which the 50 percent taxation of benefits begins.

One of the more important policy conclusions of their study related to the effects on retirement hours of eliminating the 50 percent taxation of benefits of retirees. In terms of Figure 5, this would result in line CD being extended upward and to the left.

²However, a second peak appears at age 62 even though the constraint has neither strong convexity or nonconvexity there. The explanation for this clustering given by the authors was the presence of liquidity constraints, for age 62 is the earliest age that individuals can receive benefits.

Figure 7

Distribution of earnings as a fraction of Social Security exempt amount



Source: Burtless and Moffitt (1984)

Burtless and Moffitt found that 80 percent of the retirees were at point *C*—not working at all—and 10 percent earned less than the exempt amount (i.e., they were on the first segment, *CD*). Consequently, 90 percent of the retired population would be unaffected by the elimination of the tax. For the remaining 10 percent—one-third of whom were above the kink and two-thirds at it—the authors estimated that hours of work would increase by a large amount, about 11 hours per week. But average hours in the retired population as a whole would only increase by about one hour per week, a small amount. Thus the heterogeneity of the population and its distribution over the constraint is a key determinant of the effects of the policy change.

The Statistical Problem and Existing Approaches

As noted earlier, there are two aspects of the statistical problem. One is that the response to changes in the constraint may be highly nonlinear, as individuals jump from segment to segment or stick at kinks. Second, data always show that a set of observationally-equivalent agents, when faced with an identical constraint, choose to spread themselves over different points along that constraint. They may then respond in very different ways to changes in that constraint. Therefore, it is essential that the stochastic specification of the econometric model allow one or more of the parameters in the demand function (or utility function) to vary across individuals. In other words,

it is essential that the model contain an error term for “unobserved heterogeneity” of individual preferences.³

We shall now outline four approaches to the estimation of the parameters of the demand or utility function. The first two will be the simplest but will be rejected outright; the third will be a bit more sophisticated but will require several approximations; the fourth will be the most satisfying theoretically and statistically but will also be the most difficult computationally.

Approach 1: OLS Estimation of the “Complete” Demand Function

This approach is to a great extent a straw man because no analyst has ever applied it. The approach attacks the problem head-on by applying OLS to what may be termed the “complete” demand function, by which is meant the function describing the choice of the global utility maximum as a function of all the slopes and locations of all the segments and kinks in the constraint. Econometrically, one simply regresses the amount demanded (X), or labor supply if that is the application, on all the slopes and all the intercepts of all the segments in the constraint. For example, in the Social Security example in Figure 5, there are three segments and therefore six possible variables for the regression. However, there are only three variables that determine the location of the entire constraint for each retiree in the sample— W , G , and $\hat{H}$ —so there could only be three variables in the regression.

There are two serious problems with this approach. First, the estimation of such a linear demand function obscures the nature of the nonlinearities that are at the heart of the comparative statics discussed previously. The “true” demand function is highly nonlinear and is even nondifferentiable at many points. Some small changes in some prices and incomes will result in an identically zero change in X —because they pertain to a segment the individual is not on—and other small changes will result in large, non-marginal changes in X —because the individual jumps to a different segment. In fact, in the general n -segment case allowing both convex and nonconvex kinks, the demand function may contain any number of vertical jumps, horizontal sections, forward- and backward-bending regions, and points of nondifferentiability.⁴ The constant coefficients in a linear regression force the response to be fixed regardless of initial location.⁵

³Such heterogeneity of preferences is no doubt always present, even when constraints are linear. But in the linear case, it is generally assumed that the price and income elasticities of demand are the same for all consumers—that is, that differences in preferences only affect the level, not the coefficients, of their demand functions. That cannot be assumed in the kinked constraint case because it is clear that the initial location of the individual—the initial level of demand—will often affect the type of response to a change in the constraint.

⁴Hanoch and Honig (1978) worked out a specific example of such a complete demand function for one particular budget constraint. Their graphs show how nonlinear the demand function can be in price and income.

⁵The linearized function may well predict changes in the mean value of X around the mean of the data, but extrapolation much beyond the mean would not be possible.

A second difficulty with application of OLS to the linearized function arises from the assumption of a linear error term. By tacking an error term on to the end of the equation, no variation is allowed in the coefficients across individuals and hence “unobserved heterogeneity of preferences” is not represented in the specification. Put differently, estimating such a function would result in the estimation of fixed values for the parameters of those functions for all individuals in the population. Consequently, for example, simulations conducted from the results would predict that all individuals respond identically to a change in the constraint, an undesirable feature given the importance of differences in response by different individuals discussed previously.

The underlying problem with this approach is its use of a linear regression equation to approximate a nonlinear function. This can be illustrated formally most easily for the two-segment nonconvex case graphically depicted in Figure 2. Define the direct utility function as $U(X, Y)$, the indirect utility function $V(P, M)$ as the maximum utility obtainable on a linear constraint (or segment) with slope P and intercept M , and $g(P, M)$ as the demand function resulting from utility maximization on such a segment. Then the complete demand function in the nonconvex case is the following:

$$(2) \quad X = D[g(P_1, M)] + (1 - D)[g(P_2, \tilde{M})]$$

where

$$(3) \quad D = 1 \text{ if } D^* \geq 0 \quad D = 0 \text{ if } D^* < 0$$

$$(4) \quad D^* = V(P_1, M) - V(P_2, \tilde{M})$$

Equations (3) and (4) jointly determine which segment yields maximum utility. According to (4), if D^* is positive, utility is higher on segment 1 than on segment 2. Hence, according to (3), D equals 1 and demand for the good, X , equals $g(P_1, M)$ according to (2). If D^* is negative, D equals 0 and demand for X equals $g(P_2, \tilde{M})$. This “complete” demand function, described jointly by all three equations, is highly nonlinear in the four variables determining the constraint— P_1 , P_2 , M , and $\tilde{M}$. They appear in the segment-choice equation (4), for example, where a small change in any one of them could cause no change in the variable D or a change from 1 to 0 or vice versa. This would cause a discrete, non-marginal change in X . It is not surprising that OLS with constant coefficients will inaccurately represent this function.⁶

Approach 2: OLS Estimation of the Marginal Demand Functions

A second approach is to estimate the “marginal” demand functions directly—that is, the functions that describe choice within segments. This approach was taken by

⁶ However, nonlinear estimating procedures such as nonlinear least squares and maximum likelihood may be applicable, as discussed further below.

Hall (1973), who applied OLS to the function

$$(5) \quad X_i = g(P_i, M_i) + \varepsilon_i$$

for each individual i in his sample, where X_i is the individual's observed quantity demanded, P_i is the price for the segment upon which each individual was observed to be located, M_i is the "virtual" income for that segment (i.e., the intercept $\tilde{M}$ or M in Figures 1 and 2), and where ε_i is an error term. In Hall's case, X was hours of work and the federal income tax schedule induced the nonlinearity in the constraint. This "marginal" demand function shows the amount the individual would demand if he or she were to face a completely linear constraint with slope P_i and intercept M_i .

Unfortunately, the OLS estimation of equation (5) is subject to potentially severe bias. The right-hand side variables for price and virtual income are endogenous because they are, after all, partly selected by the individual by choice of segment. Consequently, causality runs from the choice of X to the choice of price and virtual income (i.e., choice of segment), rather than vice versa.

That this endogeneity is fundamental to the problem can be seen by examining Figures 1 and 2. In Figure 1 it is clear that, on average, individuals who choose segment 1 will have low prices and virtual incomes but also low X , generating positive correlations between each of them and X . For example, in Hall's case of the federal income tax and hours of work (see Figures 3a and 3b), wage elasticities should be biased downward because those workers with low net wage rates (i.e., high marginal tax rates) must also have high hours of work by the nature of the income tax schedule. The magnitude of the bias naturally will depend upon the degree of nonlinearity of the budget set. However, it has been shown in several different applications that the bias in OLS can be extremely large, so much so as to actually reverse the signs of the estimated price and income elasticities (Dubin, 1982; Megdal, 1987; Moffitt and Nicholson, 1982).

In Figure 2 it is clear that, on average, individuals who choose segment 1 will have high prices and virtual incomes but low X , generating negative correlations between each of them and X . For example, in the case of a welfare program (Figure 3a), welfare recipients will have low labor supply and low net wages because they face high "tax" rates. These correlations would exist regardless of the magnitudes of price and income elasticities and are merely a result of the shape of the constraint. Consequently, OLS estimation of (5) may simply generate an estimate of the budget constraint.⁷

⁷Another way to describe the problem is to note that equation (5) could be estimated even if all individuals in the sample face the exact same budget constraint—that is, even if all individuals face the same market prices, have identical incomes, and face the same tax schedule with identical marginal tax rates and kink locations. But in this case it should be impossible to identify price and income elasticities because there is no true variation in price or income—there is only variation in observed marginal price and income because different individuals have different utility functions and hence choose different segments.

Another problem with this specification is that it is silent concerning which individuals, if any, locate at the kinks. In fact, the approach assumes that no individuals are at the kinks at all. Yet, Figure 1 implies that there should be a clustering of individuals at the kink because utility maximization there is compatible with many different marginal rates of substitution. Of course, in a real set of data no individuals will be located exactly at a kink—if, for example, the kink location is specified to the third decimal place—but this does not mean that there are no individuals close to the kink who are where they are, effectively, because they wish to locate “close” to a kink. Sometimes clustering is clearly revealed in the data, as in the Social Security example in Figures 6 and 7. As a general rule, the stronger visual evidence of clustering is found, the more convex is the kink, as should be expected. It may be difficult to detect clustering in the federal income tax, for example, because the kinks are not very convex—marginal tax rates do not change by 50 from one bracket to the next as they do for Social Security. In any case, the function in (5) misspecifies the demands of kink individuals by assuming that they have a tangency along a segment when they do not.

Approach 3: Estimation of Marginal Demand Functions with Instrumental Variables

The third approach to the problem attempts to correct the problem with the second approach created by reverse causality—that is, the problem that the price and virtual income are partly determined by the choice of demand or labor supply (X), and not the other way around. Problems of reverse causality are often addressed in econometrics by analogy with simultaneous equations, where reverse causality is also generally present. In the simultaneous equations case, the problem can be portrayed as arising from a correlation between a right-hand side variable and the error term in the regression equation (that is, the variable is endogenous). One solution to this problem is to construct an instrumental variable—a variable correlated with the endogenous regressor variable but uncorrelated with the error term. Likewise, in the estimation of (5), the problem of bias partly arises because the values of the error term, ϵ , are correlated with the price and virtual income (for example, those with high values of ϵ will have high hours of work and therefore low net wage rates under a progressive income tax). An econometric solution to this problem may therefore be possible if instrumental variables can be developed for the price and virtual income variables—that is, if price and virtual income in (5) can be replaced with truly exogenous variables that are correlated with them but uncorrelated with the error term.

There have been a number of attempts along this line. In two studies of labor supply (that is, hours of work), one on the effect of the federal income tax (Rosen, 1976) and one on the effect of a negative income tax (Hausman and Wise, 1976), the marginal tax rate used to construct net wage rates and virtual incomes was evaluated at the same level of hours of work for all individuals. The tax rate t in the calculation of the net wage rate $W(1 - t)$ (see Figures 3a, 3b, and 3d) was computed for each

individual as the tax rate he or she would face if working 40 hours per week. The resulting net wage variable must be uncorrelated with the error term in the labor supply equation because it is not constructed on the basis of actual, chosen hours of work.

A related approach popular in the study of the effect of the federal income tax on charitable deductions is to regress the amount of the charitable deduction taken by a taxpayer not on the actual marginal tax rate faced—that tax rate is dependent upon the amount of the deduction chosen, since a greater deduction reduces taxable income and hence moves the individual into a lower tax bracket—but on the marginal tax rate the taxpayer would face if no charitable deduction were taken. The resulting tax rate is that faced by the taxpayer for the first dollar of charitable deductions, and hence is called the “first-dollar” marginal tax rate. It qualifies as an instrumental variable because its value is not based upon the actual charitable deduction taken.

A more sophisticated procedure followed by Hausman et al. (1979) modified the Rosen and Hausman–Wise method by calculating the marginal tax rate not at the same value of X for all individuals, but rather at a predicted value of X based upon a set of exogenous characteristics. In the Hausman et al. procedure, the results of a first-stage estimation of a reduced-form equation for X were used to calculate a predicted value of X for each observation, and a marginal tax rate was computed for each individual in the sample at that predicted value. This was then used in the construction of the price and virtual income variables.⁸

All of these procedures suffer from the defect that they obviously partially misrepresent the actual situation faced by the individual. In the Rosen and Hausman–Wise procedure, for example, the actual tax rate faced by the individual may be quite different from the tax rate that would be faced at full-time work. Likewise, in the first-dollar price methodology, the actual tax rate faced by the taxpayer might deviate considerably from this price. Thus the procedures create some specification difficulties at the same time as they reduce the endogeneity problem.

A related procedure suggested by Heckman and MaCurdy (1981) permits the estimation of (5) by OLS but with the inclusion of an additional variable capturing “selection bias.” In a two-stage procedure, a segment-choice equation is first estimated and the results are used to construct a “selection bias” variable based upon the probability of segment location, which is then included in the OLS estimation of (5). The procedure is an application of more general selection-bias techniques developed earlier (Heckman, 1979; see Maddala, 1983, for a review).

While these procedures deal adequately with the endogeneity of marginal price and income, they leave the problem of kink locations unaddressed. The problem does not arise in the nonconvex model because location at the kink cannot be optimal. But

⁸Note that a direct two-stage least-squares procedure in which the price and virtual income variables are regressed upon a vector of exogenous variables for the first stage would be inefficient because the gross price and income are exogenous and do not need to be instrumented.

it would arise in the convex case. Here the complete demand function can be written:

$$(6) \quad X = D_1[g(P_1, M)] + D_2[g(P_2, \tilde{M})] + (1 - D_1 - D_2)X^*$$

where

$$(7) \quad D_1 = 1 \text{ if } D_1^* \geq 0 \quad D_1 = 0 \text{ if } D_1^* < 0$$

$$(8) \quad D_2 = 1 \text{ if } D_2^* \geq 0 \quad D_2 = 0 \text{ if } D_2^* < 0$$

$$(9) \quad D_1^* = X^* - g(P_1, M)$$

$$(10) \quad D_2^* = g(P_2, \tilde{M}) - X^*$$

Equations (9)–(10) show that the determinants of segment location do not explicitly require the utility function in this case. For example, in (9), if $g(P_1, M) < X^*$, and hence $D_1^* > 0$, the utility maximum on segment 1 lies to the left of X^* as shown by the segment-1 indifference curve in Figure 1. But if the indifference curve is tangent there, utility at the kink and on segment 2 must be lower. Hence, $D_1 = 1$ according to equation (7) and demand equals $g(P_1, M)$ according to (6). If $g(P_2, \tilde{M}) > X^*$, D_2^* is greater than 0 and $D_2 = 1$, implying that the individual is on segment 2 and has demand equal to $g(P_2, \tilde{M})$. But what if the person is located at the kink, as shown by the indifference curve at X^* in Figure 1? In this case we will have $D_1 = 0$ and $D_2 = 0$ and, according to equation (6), demand will simply equal X^* .⁹

The problem with the instrumental variable procedures is that they do not easily address the problem of kink locations. In most applications of the procedure, it is assumed that no individuals are located at kinks, creating the same type of misspecification that was discussed for Approach 2. For example, in the case of strong clustering as shown in the Social Security examples in Figures 6 and 7, it would not be appropriate to assign observations clustered at the kinks to one of the segments. Alternatively, the observations at the kinks could simply be deleted from the sample. But this would involve throwing out valuable information and, in addition, it would be difficult to do so if clustering around the kinks were not so visually obvious as in Figures 6 and 7. If clustering were not so pronounced, some arbitrary decision would have to be made regarding how close an observation has to be to a kink to be called a “kink observation” and therefore deleted from the sample. There are no solid rules for making such choices, and the rule chosen could easily affect the results.

Approach 4: Joint Estimation with Two Error Terms

In a pathbreaking paper, Burtless and Hausman (1978) solved the problems with the three previous techniques in an internally consistent, clean way that simultaneously preserves the closeness of the link between economic theory and the economet-

⁹See Moffitt (1986a) for a proof.

rics. Burtless and Hausman were interested in the labor supply effects of a negative income tax, which creates a nonconvex budget set, and of the federal income tax, which creates a convex budget set. Figure 3d illustrates the negative tax case and Figures 3a–3b illustrate the federal income tax case (though in actuality there are, of course, more than two segments).

The main innovation is to permit the econometric model to have two error terms instead of one. One error term represents unobserved heterogeneity of preferences, and is interpreted as representing variation in utility functions across individuals. For example, different individuals have different tastes for work and hence will choose to work different amounts even if they face the same budget constraint. The second error term represents “optimization error” and arises because individuals cannot fine-tune their hours of work precisely. As a result, for example, they will not locate precisely at a kink even if they desire to do so.¹⁰

The model can be illustrated with the case of the federal income tax (see Figures 3a–3b). Let H be hours of work, $W(1 - t_1)$ be the net wage on the first segment, M be its intercept (that is, “virtual income”), $W(1 - t_2)$ be the net wage on the second segment, $\tilde{M}$ be its intercept, and let the kink occur at H^* hours of work. Let the heterogeneity error term be ν and the optimization error term be ε . Then the model can be written in terms of two “equations”:

$$\begin{aligned}
 H &= \alpha + \beta W(1 - t_1) + \gamma M + \nu + \varepsilon && \text{if utility max. on segment 1} \\
 (11) \quad H &= \alpha + \beta W(1 - t_2) + \gamma \tilde{M} + \nu + \varepsilon && \text{if utility max. on segment 2} \\
 H &= H^* + \varepsilon && \text{if utility max. at the kink}
 \end{aligned}$$

Utility max. on segment 1 if:

$$\alpha + \beta W(1 - t_1) + \gamma M + \nu < H^*$$

(12) Utility max. on segment 2 if:

$$\alpha + \beta W(1 - t_2) + \gamma \tilde{M} + \nu > H^*$$

Utility max. at kink if:

$$\alpha + \beta W(1 - t_1) + \gamma M + \nu > H^*$$

$$\alpha + \beta W(1 - t_2) + \gamma \tilde{M} + \nu < H^*$$

¹⁰An alternative source of error is simple measurement error, which would arise if the data are inaccurate. Measurement error and optimization error are conceptually very different, but the second error term in the model can represent either or both—they cannot be econometrically distinguished in this model. Hausman (1985, pp. 1264–1269) provides an extensive discussion of different sources of randomness, including specification error as well.

Equation (11) shows the set of equations describing the hours of work an individual would have under each of the three utility-maximizing possibilities. If the individual maximizes utility on segment 1, H is determined by a linear regression equation containing the slope and intercept of that segment ($W(1 - t_1)$ and M) plus the two error terms—each individual will have a particular “taste” error term, ν , as well as an optimization error, ε . If maximizing on segment 2, H is determined by the corresponding net wage and virtual income on that segment. However, if the individual maximizes utility at the kink, H is simply equal to the kink value plus the optimization error, ε . No preference error term is involved because location at the kink means that desired H is at a single point.

The three “equations” in equation set (12) describe the utility-maximizing choice of segment or kink. Segment 1 is chosen if desired hours of work on segment 1 (the first bracket in Figures 3a–3b) is lower than H^* —that is, if desired labor supply is actually on that segment, and not greater than H^* . Note that the optimization error term is not involved in this decision but the preference error is—this is the key difference between equations (11) and (12). The individual desires segment 2 if hours of work on that segment are greater than H^* —that is, if desired labor supply is actually on that segment and not less than H^* . These two conditions show that individuals with high values of ν (high unobserved tastes for work) will choose segment 2 and that individuals with low values of ν (low unobserved tastes for work) will choose segment 1. The third condition, that for the kink, indicates that the kink will be desired if ν falls in between these high and low values. If ν is so high that the net wage and virtual income for segment 1 put the individual to the left of H^* , but if ν is so low that the net wage and virtual income for segment 2 put the individual to the right of H^* , the individual will choose the kink instead.

Burtless and Hausman showed that the coefficients in these equations can be estimated with the technique of maximum likelihood. In this technique, the two error terms are assumed to have particular distributions (they assumed normality for both) and then the probability of observing the value of H for each individual is derived. The parameters are then chosen by iteratively searching over all possible parameter values, ending up with the set that maximizes the product of the probabilities of observing the H values for all observations in the sample.

Estimation of the model also generates estimates of the variances of both error terms. The estimates of those variances are determined by the degree of clustering around the kinks in the data. If clustering is very strong—as in the Social Security example in Figures 6–7—the estimation procedure will generate a small estimate of the variance of ε because the clustering implies that “optimization error” is small (that is, individuals are relatively successful at locating close to the kink). On the other hand, if there is very little clustering around the kinks, the estimation procedure will generate a relatively large value of the variance of ε .

The model can also be estimated if either of the error terms is omitted. If only the optimization error is present, for example, ν does not appear in equations (11)–(12). This model was first estimated by Wales and Woodland (1979) in a study of the effect of the income tax on labor supply. The disadvantage of this model is that no

preference heterogeneity is allowed, and therefore it is assumed that all individuals with the same budget constraint are maximizing utility at the same point—that is, it is assumed that all the spread of the observations over the budget constraint is the result of optimization error, not variation in tastes for work.¹¹ On the other hand, if only heterogeneity error is present, the model would imply that individuals should locate exactly at the kink points, with no error. In the Social Security example described in Section II, the authors made this assumption because the clustering of retirement ages and post-retirement hours of work were so strong that individuals were “almost” locating exactly at kink points. For the retirement age (see Figure 6), the authors rounded to the nearest age (age 65 for anyone retiring within six months of their 65th birthday) and for hours of work (see Figure 7), the authors assigned anyone within two hours of their respective kink points to the kink. The disadvantage of this procedure is that it would be difficult to implement if the degree of clustering were not so pronounced. See Hausman (1980), Zabalza et al. (1980), and Moffitt and Nicholson (1982) for other examples of heterogeneity-only models.

An unfortunate but necessary implication of the model is that we (the analysts) cannot take observed segment locations as utility-maximizing locations. The presence of the optimization error term implies that an individual observed to be located on any segment of the constraint may in fact have a utility maximum on a different segment. As shown in equation (11), the error term ε may move H to a different segment. Consequently, if this model is correct, all three approaches described previously should generate biased and inconsistent estimates because they incorrectly assume that observed segment locations are utility-maximizing locations—that is, they assume that the price and virtual income on the observed segment are those that enter the demand function and hence determine demand.

Current Issues Involving Kinked Budget Constraints

Four issues under current discussion in the literature are noted in this section. First, the literature on the estimation of demand functions subject to kinked budget constraints has become concerned with the more general issue of flexible functional forms for demand. To date, most of the studies have assumed utility functions that are guaranteed to be globally concave over easily defined ranges of the parameters (like the Stone–Geary utility function) or have forced the coefficients in their labor supply or demand functions to be consistent with indifference curves of the “correct” shape (downward sloping, and so on). As in the consumer demand literature in general, it would be more desirable here to use more flexible and less restrictive functional forms—such as the translog utility function, for example. Unfortunately, this is difficult to do in the kinked budget constraint model because the model requires not just that the

¹¹The optimization-error-only model is essentially the model described in Approach 1. However, rather than apply OLS to estimate the model, the nonlinearities require the use of nonlinear least squares or maximum likelihood.

indifference curves be of the correct shape in the neighborhood of the observed value of, say, hours of work, but also of the correct shape everywhere else on the constraint. This requirement arises because the presence of optimization error implies that true utility maximization is not necessarily at the observed point. Unfortunately, most flexible functional forms for utility such as the translog cannot guarantee global concavity for all ranges of the data. The practical problem of how to permit more flexibility in the kinked budget constraint model has not yet been fully worked out.¹²

Second, there is a concern that the budget constraint is not accurately measured in some cases and that a special type of errors-in-variables bias results (Heckman, 1983). To estimate the model requires that the analyst determine the exact locations of all kinks in the constraint for each individual in the sample. However, for example, it is difficult given the data available to accurately determine the exact location of kink points in the federal income tax for each individual in a household survey because insufficient data are available on deductions, filing status, number of exemptions, itemization status, tax avoidance, and so on. As a result, the locations of the kinks assumed for the analysis may be inaccurate. This could be an alternative explanation for the lack of visual clustering around convex kink points in some data plots. This is a difficult problem to address because the requisite data are essentially not available, and its empirical importance is consequently not yet clear. However, Triest (1987a) conducted some Monte Carlo experiments on the issue and concluded that the magnitude of the bias from the problem is not likely to be large. In addition, in a separate paper on the income tax (Triest, 1987b), he studied explicitly the endogeneity of deductions, perhaps the largest potential source of error in budget constraint determination in prior studies.

Third, it has been suggested that smoothing a piecewise-linear constraint into one that is everywhere differentiable may permit simpler estimating techniques to be used. By eliminating the kinks, smoothing the constraint would eliminate the need to identify observation at kinks and hence would reduce the force of one of the main objections to the method of instrumental variables estimation of the marginal demand functions as in Approach 3. It is possible as well that a single error-term model could then be used. Since smoothing techniques have not yet been tested in any systematic fashion, it remains to be seen whether they will prove useful.¹³

Fourth, there has been a concern that the normality assumption for the two error terms in the model is overly restrictive and may result in inconsistent estimates if the true distributions are not normal. Although little work has been completed on this problem, it clearly calls for more specification testing of the distributional assumptions, at minimum, or, preferably, the use of less parametric estimating methods.

¹²See Kapteyn et al. (1986) and MaCurdy et al. (1988) for studies of this issue.

¹³See Bourguignon and Magnac (1988) and MaCurdy et al. (1988) for some preliminary attempts at smoothing. The success of the method depends on the number of kinks and their convexity, and on the accuracy with which the true constraint can be approximated. Note that smoothing the constraint does not eliminate the problem of nonconvexities in the constraint or the problem of imposing global concavity of the utility function.

Summary

The development of econometric methods for the analysis of the effects of kinked budget constraints has arisen over the 1970s and 1980s in direct response to the need to analyze the incentive effects of government tax and transfer programs. As the discussion in this paper has attempted to demonstrate, such incentive effects are quite complex and often counterintuitive, and their estimation requires careful consideration of the variation in preferences across different individuals as well as the sources of other variation in the data. Many of the simple methods that are applicable in other areas of applied econometrics do not work here; the detailed specification of individual responses in the econometric procedure is absolutely necessary.

This point becomes clear whenever the results of an estimated model are used to simulate the effects of policy changes. For example, the study of the effect of the income tax on labor supply by Hausman (1981) has by far had the most influence on policy of any study in this area, for his results and subsequent simulations of tax reform were extensively discussed in the tax reform debates of the 1980s. The attractiveness of the model lay in its ability to simulate the effects of the very detailed changes in the bracket structure of the income tax that were under consideration and which were implemented in the Tax Reform Act of 1986, taking into account both the number of individuals affected in each bracket and the degree to which they would change brackets after a change. It would have been essentially impossible to accurately forecast the effects of the Act on labor supply without a kinked-budget-constraint model of the general type discussed in this paper.

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